SATHISH RAJA

Senior Full Stack Engineer | Solution Engineer | Software Tech Lead | AI Full Stack Engineer (MERN Stack – Node.js, React.js/Next.js, TypeScript)

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer / Software Tech Lead with 14+ years of progressive experience in full stack web application development, solution engineering, and AI-agent architectures across Singapore. Experienced across Full Stack Engineer, Senior Software Engineer, Solution Engineer, Web Application Developer, Application Developer, Programmer, and Analyst-facing responsibilities, translating business requirements into scalable software solutions. Proven track record designing distributed task-processing queues, optimizing Docker containerization to cut deployment errors by 50%, increasing API transaction throughput by 40%, and delivering 10+ production-grade applications across government, finance, robotics, and SaaS domains using the MERN stack (MongoDB, Express.js, React.js, Node.js) and TypeScript. Skilled in Node.js, React.js/Next.js, TypeScript, and modern GenAI/AI Full Stack frameworks including LangGraph, Retrieval-Augmented Generation (RAG), and the Model Context Protocol (MCP). Recent hands-on focus includes building and deploying SmartHire AI, a live production AI recruitment platform (smarthire.sathishraja.com) powered by Google Gemini, LangGraph, LangChain, Qdrant Cloud vector search, RAG-style semantic retrieval, BullMQ/Redis, MCP tools, role-based access control (RBAC), and human-in-the-loop (HITL) approval workflows.

CORE COMPETENCIES

Full Stack Development | MERN Stack Engineering | Solution Engineering & Solution Architecture | Software Tech Lead / Technical Leadership | Web Application Development | AI Full Stack Development | System Design & Architecture | Requirements & Business Analysis | Software Development Life Cycle (SDLC) | Agile / Scrum Methodology | API Design & Integration | Microservices Architecture | Database Design & Optimization | Cloud Solution Deployment | Code Review & Mentoring | Cross-Functional & Stakeholder Collaboration | Client-Facing Technical Consulting | Programming & Debugging | Performance Optimization | DevOps & CI/CD

CORE TECHNICAL SKILLS

AI & LLM Architecture: Google Gemini, LangGraph, Multi-Agent Orchestration, Agentic AI, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP SDK), Human-in-the-Loop (HITL), Prompt Engineering, Vector Embeddings, Semantic Search, Qdrant Vector Database / Qdrant Cloud, Ollama, LangChain

Frontend Engineering: Next.js 16, React 19, TypeScript, Zustand, TanStack React Query, Tailwind CSS v4, shadcn/ui, Radix UI, React Hook Form, Framer Motion, Redux Toolkit

Backend & Systems: Node.js, Express 5, BullMQ, Redis (ioredis), Mongoose, RESTful APIs, Microservices, RBAC, System Design

Databases & DevOps: MongoDB Atlas, Qdrant Cloud, MySQL, PostgreSQL, Docker, AWS (EC2, S3), Vercel, Render, Nginx, CI/CD Pipelines (GitHub, Bitbucket)

Security & Utilities: JWT, Helmet, CORS, Express Rate Limit, Bcrypt, Zod Validation, Multer, Cloudinary, Resend, Centralized Error Handling, Automated Unit Testing (TDD)

PROFESSIONAL EXPERIENCE

Principal Full-Stack & AI Consultant (Independent) – SmartHireAI Platform Project

Independent Consulting | 10/2024 – Present | Singapore / Remote

- Architected and engineered an end-to-end, multi-tenant AI Applicant Tracking System (ATS) with secure routing dashboards for candidates, admins, HR recruiters, and hiring managers, including public careers workflows, JWT authentication, and role-based access control (RBAC). Live at smarthire.sathishraja.com.
- Developed complex, non-linear recruiting workflows using @langchain/langgraph and @langchain/ollama to coordinate autonomous AI screening agents; orchestrated the AI Recruiter with LangGraph to retrieve candidates, evaluate skills/experience/relevance, rank recommendations, propose actions, and pause for human approval before side effects (human-in-the-loop).
- Built a heavy-duty document parsing pipeline using pdf2json and pdfjs-dist to ingest candidate resumes, running asynchronous BullMQ/Redis background jobs for PDF extraction, Google Gemini profile analysis, and scoring.
- Implemented semantic candidate retrieval with Google Gemini embeddings and a Qdrant Cloud vector database, including jobId-filtered vector search for job-specific candidate matching via RAG-style semantic retrieval.

- Implemented an asynchronous background worker cluster using BullMQ and Redis (ioredis) to handle long-running profile screening and ML candidate ranking without blocking the core API event loop.
- Shipped a highly interactive frontend using Next.js 16 (Turbopack), React 19, Zustand, Tailwind CSS v4, shadcn/ui and Radix UI components, featuring real-time data fetching with TanStack Query and interactive Kanban boards via dnd-kit; deployed the frontend on Vercel and the API/worker on Render with custom domains.
- Deployed the Model Context Protocol (MCP SDK) for create/reschedule/cancel interview and candidate email actions, enabling autonomous agents to safely read file states, trigger calendar tools, and dispatch high-deliverability candidate notifications via Resend and Nodemailer, with human-in-the-loop approval and duplicate-interview protection.
- Integrated MongoDB Atlas, Qdrant Cloud, managed Redis, Cloudinary resume storage, Resend email delivery, Helmet, CORS controls, request validation, and centralized error handling across the platform.

CodeSense AI - Multi-Agent AI Code Review Platform

Independent Project | Singapore / Remote

- Built and deployed a GitHub-integrated multi-agent AI platform that automatically reviews pull requests using specialized security, performance, code-quality, and testing agents, publishing contextual inline feedback through the GitHub Review API.
- Designed an event-driven asynchronous review pipeline using signed GitHub webhooks, Express, BullMQ, Redis, and MongoDB with job retries, persistent review state, and publication checkpointing.
- Implemented a deterministic coordinator that combines agent output, deduplicates overlapping findings, and prioritizes findings by severity before GitHub publication, avoiding an unnecessary additional LLM coordination call.
- Implemented GitHub OAuth authentication, signed HTTP-only sessions, GitHub App onboarding, repository synchronization, and per-user repository authorization for repositories, pull requests, reviews, and findings.
- Built a Next.js dashboard for repositories, pull requests, reviews, and findings and deployed the production application with Vercel, Render, MongoDB Atlas, managed Redis, and custom domains. Live: codesense.sathishraja.com

Senior Software Engineer

Konica Minolta Business Solutions Asia Pte Ltd | 12/2022 – 09/2024 | Singapore

- Developed and integrated high-scale enterprise full-stack software bridging internal engineering systems and international third-party hardware modules.
- Orchestrated containerized application pipelines using Docker, standardizing deployment environments and cutting staging-to-production deployment errors by 50%.
- Designed and implemented highly performant RESTful APIs for critical automated workflows, boosting overall data transaction throughput by 40%.
- Partnered with Business Development teams, analysts, and end-clients to translate enterprise requirements into modular software components and microservice parameters.
- Coordinated with cross-functional engineering teams to evaluate and enhance software-hardware interfaces for AMR systems, analyzing proposed technical solutions against customer requirements.
- Led code interfaces and software logic configuration for Autonomous Mobile Robots (AMRs), including MiR Robots and MiR Fleet allocation, monitoring, and mission scheduling ecosystems.

Senior Software Consultant

Helius Technologies Pte Ltd | 02/2022 – 12/2022 | Singapore

- Shipped scalable enterprise web applications leveraging the Node.js ecosystem and contemporary client-side frameworks.
- Assured full backward-compatibility across rolling updates of feature rollouts on legacy code segments.
- Moderated tech-stack roadmap design reviews, driving strategic infrastructure decisions alongside engineering leaders.
- Delivered advanced technical support to resolve complex, multi-system integration challenges across project workstreams.

Senior Software Engineer

Unisoft Infotech Pte Ltd | 02/2021 – 01/2022 | Singapore

- Directed end-to-end development of client-facing platforms using Node.js backend patterns and modular React structures.
- Optimized database indexing routines and server processing pipelines to maintain low-latency responses for external consumer interfaces.
- Drove Agile sprint planning and retrospectives with the project manager, improving team velocity and delivery predictability.

Application Programmer

Optimum Solutions Pte Ltd | 08/2020 – 01/2021 | Singapore

- Built robust web interfaces for public-sector platforms, tuning UI components to pass rigorous government usability metrics.
- Developed RESTful APIs for a government initiative and integrated the eNETS payment gateway into a citizen-facing chatbot, improving transaction reliability.

Application Programmer

Housing & Development Board (via Talentvis) | 07/2017 – 08/2020 | Singapore

- Engineered the "Parking Auntie" chatbot platform, aggregating real-time geographic parking data across LTA, URA, and HDB sources.
- Implemented multi-tier photo-parsing routines enabling citizens to submit geographic photo notifications for automated verification.

EARLIER EXPERIENCE

Software Engineer – Cromly Pte Ltd, Singapore | 04/2016 – 03/2017 — Developed and customized web applications using Node.js and Koa.js for a social media design portal, marking the transition to a full Node.js-based stack; built API-centric, modular content infrastructure enabling interactions between home décor merchants and residential property owners, deployed on AWS EC2.

Software Engineer – Aspro Technologies, India | 09/2015 – 03/2016 — Managed core relational schema optimizations using clean OOP principles across full-stack software applications.

Web Application Developer – iClick Media Pte Ltd, Singapore | 11/2014 – 07/2015 — Programmed, debugged, and optimized corporate web properties powered by transactional database structures.

PHP Developer – Dons Infomedia, India | 12/2012 – 10/2014 — Authored clean backend software patterns to resolve multi-tenant business process dependencies.

PHP Developer – Nangil Infotech, India | 02/2009 – 11/2012 — Maintained consumer-facing portals using standard relational architectures and clean layout formats.

KEY PROJECT HIGHLIGHTS

- SmartHire AI: Live agentic recruitment platform built with Google Gemini, LangGraph, LangChain, Qdrant Cloud, BullMQ/Redis, MCP tools, and HITL approvals, on a Next.js and Node.js stack. Live: smarthire.sathishraja.com
- CodeSense AI: Built and deployed a GitHub-integrated multi-agent AI code review platform using Google Gemini, GitHub Apps, BullMQ/Redis, MongoDB, Next.js, and Node.js. Specialized security, performance, code-quality, and testing agents analyze pull requests; a deterministic coordinator deduplicates and severity-prioritizes findings before publishing contextual inline comments through the GitHub Review API. Includes GitHub OAuth, signed sessions, GitHub App onboarding, repository synchronization, and per-user repository authorization. Live: codesense.sathishraja.com
- sWorkflow (Konica Minolta): Built a configurable automation platform orchestrating complex workflow scripts for autonomous robotics fleets, enabling seamless third-party device integration.
- sInfra (Konica Minolta): Engineered a multi-floor operations manager integrating elevator signals with robotic fleet routing APIs, with configurable mission scheduling.
- sConnect (Konica Minolta): Developed a real-time MiR robot monitoring application with graphical performance dashboards, giving operations teams live visibility for decision-making.
- Insurance Policy Automation: Scripted custom workflow schedules that minimized clerical processing operations for corporate finance systems.

EDUCATION

Bachelor of Engineering in Computer Science & Engineering
Anna University, Chennai, Tamil Nadu, India | 06/2004 – 04/2008